

Brute force and dictionary attacks

Learning objectives

- **Differentiate between brute force and dictionary attacks**
- **Identify real-world risks associated with weak passwords**
- **Apply effective defence strategies**

Introduction to password attacks

- **Passwords are frequent targets in cyberattacks**
- **Brute force and dictionary attacks are common methods**
- **They use different strategies to guess a valid password or key**

What is a brute force attack?

- **Tries every possible character combination**
- **Guaranteed to work eventually, but can be slow**
- **Longer and complex passwords increase cracking time**

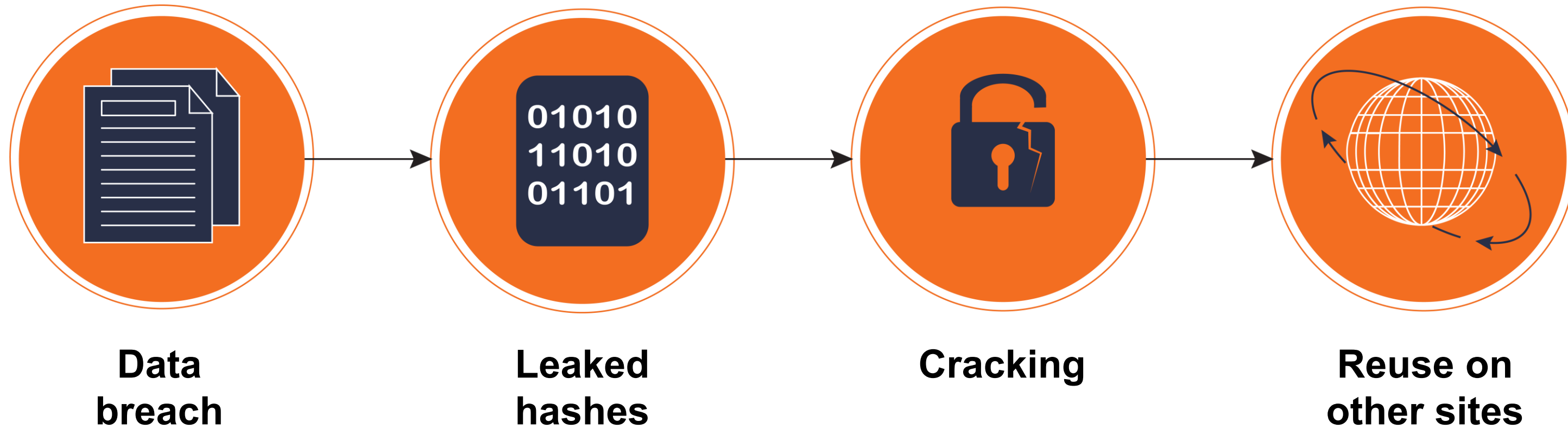
What is a dictionary attack?

- **Faster than brute force for weak passwords**
- **Uses a list of common words or phrases**
- **Targets predictable user habits**

Tools and techniques used

- **Hydra, John the Ripper and Hashcat are common tools**
- **Support both brute force and dictionary attacks**
- **Cloud computing and rainbow tables increase speed**

Real-world examples



Defences against brute force attacks

- **Use rate limiting, account lockouts and CAPTCHA**
- **CAPTCHA can slow automated attacks**
- **Strong passwords with high entropy reduce risk**

Defences against dictionary attacks

- **Avoid dictionary words and common patterns**
- **Enforce complex password policies**
- **Use password managers**
- **Store passwords with salted hashes like bcrypt or Argon2**

The importance of password hashing

- **Hashing protects passwords even if stolen**
- **Salting prevents use of rainbow tables**
- **Bcrypt and Argon2 are secure slow hash functions**

Summary

- **Brute force and dictionary attacks guess passwords and encryption keys**
- **Strong defences include strong passwords and secure hashing methods**

Next

- **Learn about man-in-the-middle attacks**